

Vinayak Jain

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Academics

Thapar Institute of Engineering and Technology, Patiala

B.Tech, Electrical and Computer Science – CGPA: 8.00

2023 – 2027

ESS ESS Convent Sr. Sec. School, Agra

CBSE (Class XII) – 83.5%

2021

Projects

Share Note — JavaScript, Node.js, Express, MongoDB, bcrypt.js, JWT

[Link](#)

- Designed and deployed **RESTful APIs** with **JWT authentication** for secure note-sharing workflows.
- Built a **scalable backend** supporting 100+ concurrent users using async handling and optimized queries.
- Integrated an **LLM-inspired NLP pipeline** to generate context-aware summaries from shared notes.
- Developed a **modular backend architecture** with middleware, routing, and reusable services.

Matrix — Backend Observability & Monitoring Service (Go), PostgreSQL

[Link](#)

- Built a **real-time monitoring system** with WebSocket dashboards for latency, traffic, and uptime tracking.
- Designed **high-throughput pipelines** using goroutines, worker pools, and asynchronous processing.
- Optimized **PostgreSQL queries** with indexing and batching to improve performance under load.
- Implemented **reliability features** including health checks, rate limiting, and system monitoring.

PulseMetrics — Go, Backend Systems, Concurrency

[Link](#)

- Engineered a **high-throughput ingestion service** handling 10K+ events/sec using non-blocking I/O and async processing.
- Implemented **batching, buffering, and worker-pool concurrency** to reduce database load by 80%.
- Designed a **scalable architecture** decoupling ingestion from persistence to improve system resilience.

Orion AI — Python, FastAPI, OpenAI API, Redis

[Link](#)

- Built a **FastAPI-based backend** to orchestrate AI agents for multi-step task execution.
- Integrated **LLM APIs** for reasoning, tool-calling, and dynamic response generation.
- Implemented **async endpoints and background tasks** to handle long-running operations efficiently.
- Designed a **modular agent pipeline** using prompt engineering and task chaining.

Coursework

Data Structures and Algorithms, Operating Systems, Database Management Systems (SQL, Query Optimization), Computer Networks, Object-Oriented Programming, System Design And Distributed Systems, Machine Learning

Skills

Programming Languages: C, C++, JavaScript, HTML/CSS, Basics of Go (Golang), Python (asyncio, OOP)

Core Technologies: React.js, Node.js, Express.js, MongoDB, PostgreSQL, MySQL

Libraries & Frameworks: WebSocket, Socket.IO, Tailwind CSS, Framer Motion, FastAPI

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker (basic), Git

Core Concepts: REST APIs, System Design, Async Programming, JWT, OAuth 2.0, Caching, Rate Limiting

Databases: SQL (joins, indexing, query optimization), MongoDB, Redis

AI & Tools: OpenAI API, Prompt Engineering, LLM Basics

Achievements

- **Qualified for Round 2 of the Adobe India Hackathon 2025** — shortlisted to build the **core system** for intelligent document understanding and semantic linking, aligned with the challenge brief. [Link](#)
- **1st Runner-up at Saturnalia Hackathon 2024** among 1,000+ participants.
- **Mentor at GirlScript Summer of Code (GSSoC) 2025.**

Volunteer

Urja – Annual Techno-Cultural Fest, Thapar University

Apr 2024 – Jun 2025

Executive Member

- Developed the **Urja website** with registration and **payment gateway** integration.

[Link](#)